Flextherm Eco G2 D

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ENG Wiring & Settings (FTE-HP)

hydronic flow control
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1. Introduction

1.1. General

The following instructions provide guidance for the installer and end user of Flextherm Eco G2 D Heat Batteries when used with compatible Heat Pumps.

These instructions must be read in conjunction with the Installation and User Instructions Manual for the Flextherm Eco G2 D product.

All installations must be carried out by a competent installer in accordance with local codes and regulations for plumbing, electrical installations and potable water supply

1.2. Symbols used

In these instructions the following symbols are being used to draw the user's attention to information of particular importance.

Warning
Indicates a hazardous situation that, if not avoided, could result in death or serious injury.

Caution
Indicates a hazardous situation that, if not avoided, could result in minor or moderate injury or material damage.

Notice
Signals information that is considered important but not hazard related.

1.3. Abbreviations

The following abbreviations are used throughout the manual:
- AC - Alternating Current
- dT - delta T (change in Temperature)
- DHW - Domestic Hot Water
- HP - Heat Pump
- kW - Kilo Watt
- PCB - Printed Circuit Board
- PVC - Polyvinyl Chloride

2. Safety

2.1. General Safety Notices

Warning
Only competent persons suitably qualified to carry out plumbing and electrical work may undertake installations, repairs or relocations of the appliance. Product training on the full range of Flextherm Eco G2 Heat Batteries is available from Aalberts hfc or authorised training partners

Warning
Risk of electric shock - potential dual supply. Always isolate the power supplies to the Heat Battery controller before working on the Heat Battery

Caution
These instructions must be read in conjunction with the Installation and User Instructions for the Flextherm Eco G2 D products

Caution
Do not operate the immersion heater until all heat exchanger circuits have been filled and the plumbing has been appropriately commissioned

3. Heat Pump sizing guideline

This section details the sizing guidance for the Flextherm Eco G2 D range of products with Heat Pumps. Its purpose is to assist and offer a guideline for choosing the correct size of Flextherm Eco G2 D Heat Battery to Heat Pump Capacity and promote sizing.
The graph below details the relationship between Flow Rate (l/min) and Power (kW) at dT's of 5 and 7, which are the most cornmon range of operation for heat pumps used in Domestic Hot Water (DHW) mode.

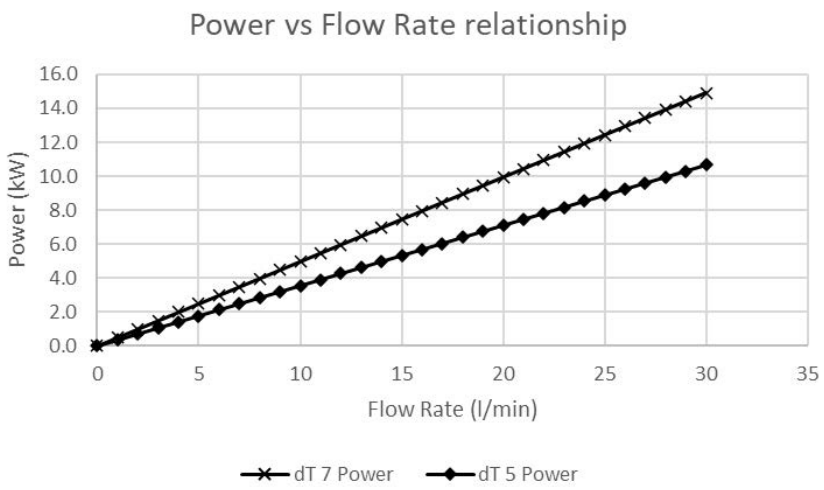

Figure 1: Power vs Flow rate relationship

The Flextherm Eco G2 D Heat Batteries work to an ideal recommended flow rate for charging via Heat Pumps as detailed in the Installation Manual (Flextherm Eco G2 D - Table 1). Therefore, the following compatibility table can be used as a sizing guide between Heat Pump Capacity and Heat Battery size:

	Heat Pump Capacity (kW)			
Heat Battery Size	**(3 to 5)**	**(5.5 to 7.5)**	**(8 to 10.5)**	**(11 to 14)**
Flextherm Eco G2 6D	O	O	O	Δ
Flextherm Eco G2 9D	!	O	O	Δ
Flextherm Eco G2 12D	!	!	O	O
! Caution:	Special consideration must be given to heat up and reheat times when combining low powered heat pumps with high capacity heat batteries			
O	fully compatible sizing			
Δ	compatible with the use of an Autobypass valve to ensure the flow rate of the heat pimp is within the recommended flow rate for the Heat Battery sizing			

Table 1: Heat Battery size and Heat Pump capacity compatibility

The Autobypass valve is always recommended in installations with Flextherm Eco G2 D Heat Batteries as it also assists in the Heat Pump's defrost condition requirements.

4. Bosch Compress 5800i AW R290 Heat Pump

Warning
All electrical wiring must be carried out by a competent person and be in accordance with the latest wiring codes and regulations.
Risk of electric shock - potential dual-supply. Always isolate the power supplies to the Heat battery, Heat Pump & Solar Diverter controller (if used) before working on the appliances.

When installing the Flextherm Eco G2 D with the heat pump, the steps below should be followed.

These settings are applicable to the following products:

- Flextherm Eco G2 D with the «Bosch HP-key» & «Bosch HP+PV-key» (Flextherm Eco G2 D manual - Section 6.4.2 for wiring instructions & Figures 2 & 3 below)
- Heat Pumps from the Bosch Compress 5800i AW R290 range.

4.1. Wiring

Using a 2-core shielded cable 0.75mm2, the cable will act as a hot water heating tank sensor from the Flextherm Eco Board - terminal J3 into the Heat Pump Indoor Unit controller PCB Terminal - (TW1 - DHW Storage Temperature) (please refer to HP installation manual). Please run the wire into the heat battery appliance via the appliance case cabling grommets and then into the control box housing through the hole available. Secure the cables in terminal J3 on the Flextherm Eco Board; please see Figures 2 & 3 below below for reference.

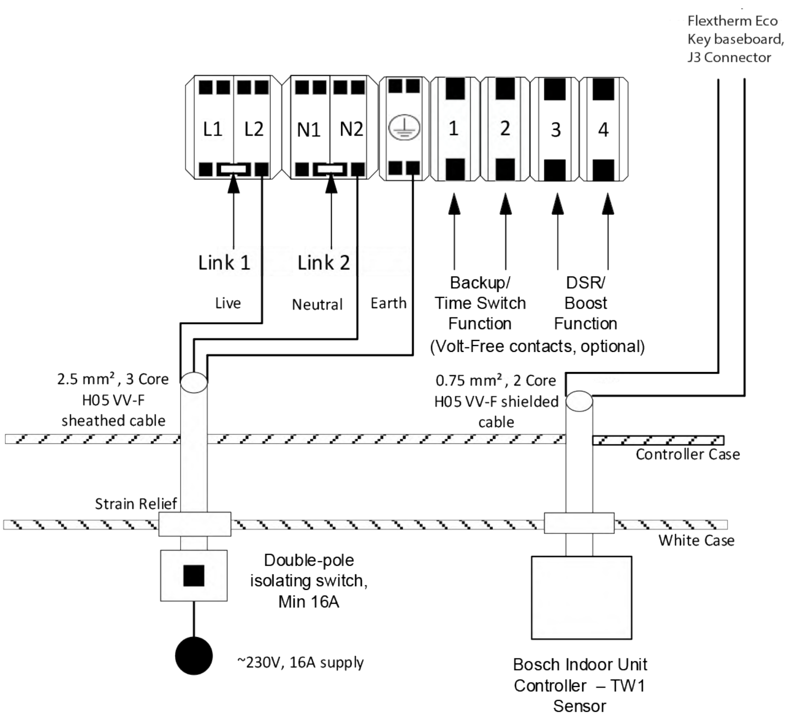

Figure 2: Flextherm Eco G2 D with Bosch HP-Key

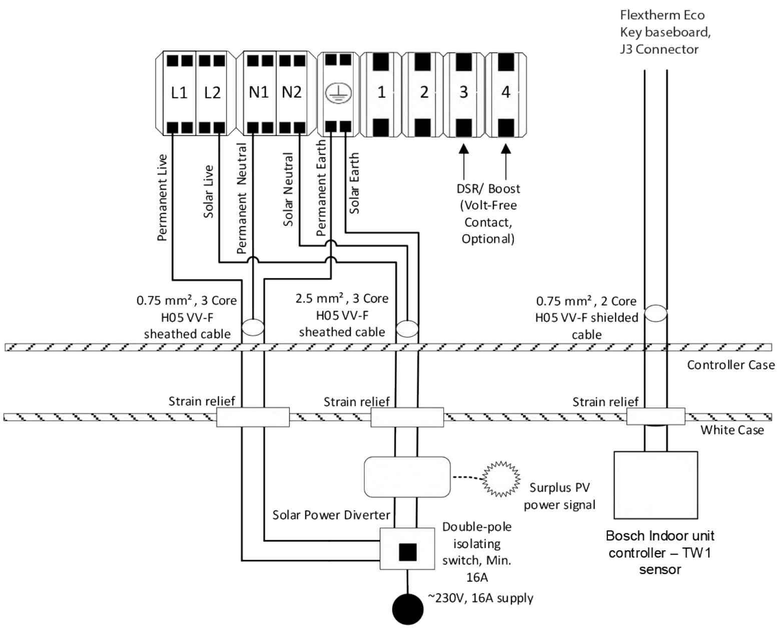

Figure 3: Flextherm Eco G2 D with Bosch PV+HP-Key

Notice
For other hydraulic connections and wiring instructions, please follow the instructions of the Flextherm Eco G2 D Installation Manual (https://flamco.aalberts-hfc.com/ex-en/docfinder).

4.2. Bosch Heat Pump Controller Settings

In the Main menu on of the Indoor Unit controller, apply the following settings to the functions listed below

Function	Settings
Aux Heater Block	On
PW2 Circulation Pump Installed	Off
DHW Operation Mode	Comfort
Comfort Start Temp	60°C
Comfort Stop Temp	65°C
Comfort temp. difference for charging	13K
Thermal Disinfection	Off
Daily Heat Up	Off

Table 2: Flextherm Eco G2 D with Bosch HP-Key settings

Please note the above are the settings to be applied to the heat pump controller itself.

Notice
If following a DHW daily timed schedule, please ensure that a minimum time window inline with the installed Heat Battery and Heat Pump capacities is chosen, and that potential defrost cycles during this window are accounted for

5. Daikin 3H HT R32 Heat Pump

Warning
All Electrical wiring must be carried out by a competent person and be in accordance with the latest local wiring codes and regulations.
Risk of electric shock - potential dual-supply. Always isolate the power supply/ies to the Heat Battery, Heat Pump & Solar Diverter controller (if used) before working on the appliances

When installing the Flextherm Eco G2 D with the heat pump, the steps below should be followed.

These settings are applicable to the following products:

- Flextherm Eco G2 D with the «Daikin HP-key» & «Daikin HP+PV-key» (Flextherm Eco G2 D manual - Section 6.4.2 for wiring instructions & Figures 4 & 5 below)
- Heat Pumps from the Daikin Altherma 3H HT range (R32)

5.1. Wiring

Using the Daikin Tank sensor part (C2294) supplied in optional kit A1135, the cable (C2294) will act as a hot water heating tank sensor from the Flextherm Eco key baseboard - terminal J3 into the Heat Pump Interface controller PCB Terminal - (DHW Tank sensor - X9A) terminal (please refer to HP installation manual). Please run the wire into the heat battery appliance via the appliance case cabling grommets and then into the control box housing through the hole available. Secure the cables in terminals J3 on the Flextherm Eco Board, please see Figures 4 & 5 below for reference.

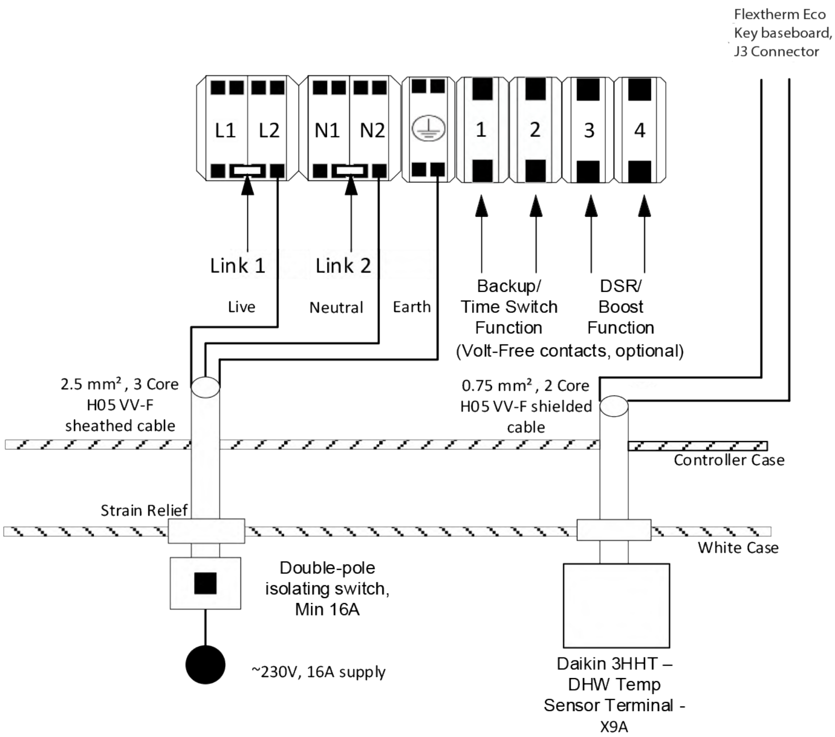

Figure 4: Flextherm Eco G2 D with Daikin HP-Key

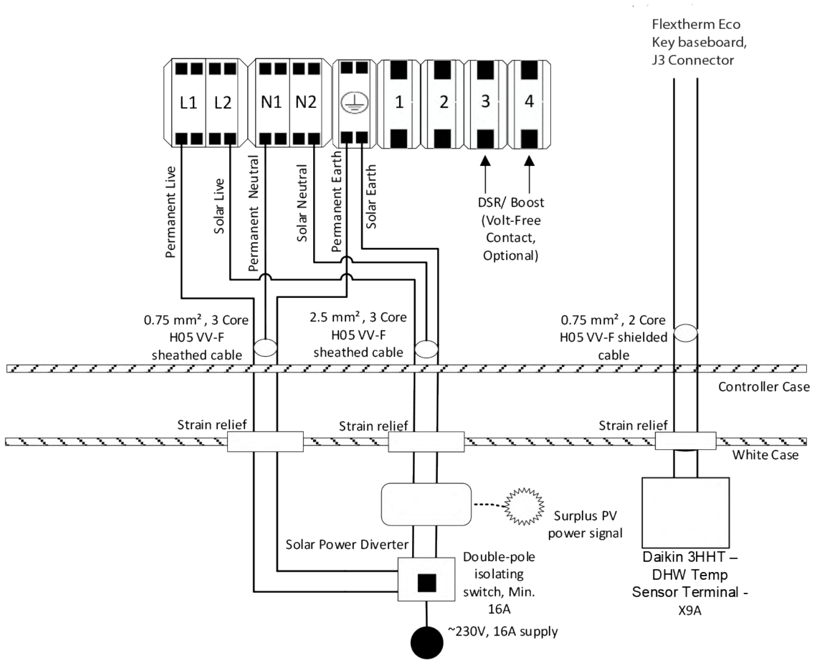

Figure 5: Flextherm Eco G2 D with Daikin PV+HP-Key

Notice
For other hydraulic connections and wiring instructions, please follow the instructions of the Flextherm Eco G2 D Installation Manual (https://flamco.aalberts-hfc.com)

5.2. Daikin 3H HT Heat Pump Controller Settings

In the Main menu on of the Hydromodule Controller, apply the following settings to the functions listed below:

Function	Settings
Tank mode:	On
DHW:	EKHWS/E Tank with booster heater installed at the side of the tank.
Emergency:	Auto SH reduced/ DHW off
Setpoint mode:	Fixed
Disinfection:	Disabled
Heat up mode:	Schedule Plus reheat (define this with end user, it is important to allocate a daily time window of 2 hours minimum if using schedule only)
Comfort setpoint:	69°C
Reheat setpoint:	50°C
DHW Hysteresis:	5°C
Target dT (Flow/ Return)	5°C

Table 3: Daikin 3H HT HP Hydromodule controller Main settings to apply

In the Field setting Menu on the Hydromodule controller, apply the following settings to the functions listed below:

Function	Settings
DHW Maximum setpoint [6-0E]	70°C

Table 4: Daikin 3H HT HP Hydromodule controller Field settings to apply

5.3. Wiring applied to Daikin Hydromodule

- Jumper Terminal 10 to 11a on terminal bed X2M, using a 0.75mm^2 wire

5.4. Using Daikin Kit A1135

Wire the 2-core PVC insulated cable provided (C2293) from the Daikin Altherma 3H HT Hydromodule Controller Booster Heater terminal connectors «X13A» (please refer to heat pump manual, run the wire into the relay box (C2291) provided into «TRIGGER INPUT AC» terminals (please refer to relay box instruction sheet). Then wire another 2-core PVC insulated cable provided (C2295) from the relay box terminals «NO1» & «C1» to the Heat Battery, into the control box housing through the opening available. Secure the cables in Terminal 1 & 2 independently. Please note that the polarity of the wires is not important in this wiring setup. Please ensure to use the provided relay backbox (C2292) & 2 x cable grommets (C2296) when running the wires into the relay box.

Notice
This function allows the backup heating element inside the Heat Battery to be activated. Please note that this will stop the Heat Battery being charged in Heat Pump mode. This can lead to increased electricity consumption, resulting in higher energy costs. This should be explained to the end user.

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6. Ecoforest ecoAIR & ecoGEO Pro R290 Heat Pump

Warning
All Electrical wiring must be carried out by a competent person and be in accordance with the latest local wiring codes and regulations.
Risk of electric shock - potential dual-supply. Always isolate the power supply/ies to the Heat Battery, Heat Pump & Solar Diverter controller (if used) before working on the appliances

When installing the Flextherm Eco G2 D with the heat pump, the steps below should be followed.

These settings are applicable to the following products:

- Flextherm Eco G2 D with the «Ecoforest HP-key» & «Ecoforest HP+PV-key» (Flextherm Eco G2 D manual - Section 6.4.2 for wiring instructions & Figures 6 & 7 below)
- Heat Pumps from the Ecoforest ecoGEO & ecoAIR PRO range (R290)

6.1. Wiring

Using a 2-core shielded cable 0.75mm2, the cable will act as a hot water heating tank sensor from the Flextherm Eco Board - terminal J3 into the Heat Pump controller PCB Terminal - (AI1 - DHW Storage Temperature) (please refer to HP installation manual). Please run the wire into the heat battery appliance via the appliance case cabling grommets and then into the control box housing through the hole available. Secure the cables in terminals J3 on the Flextherm Eco Board; please see Figures 6 & 7 below for reference.

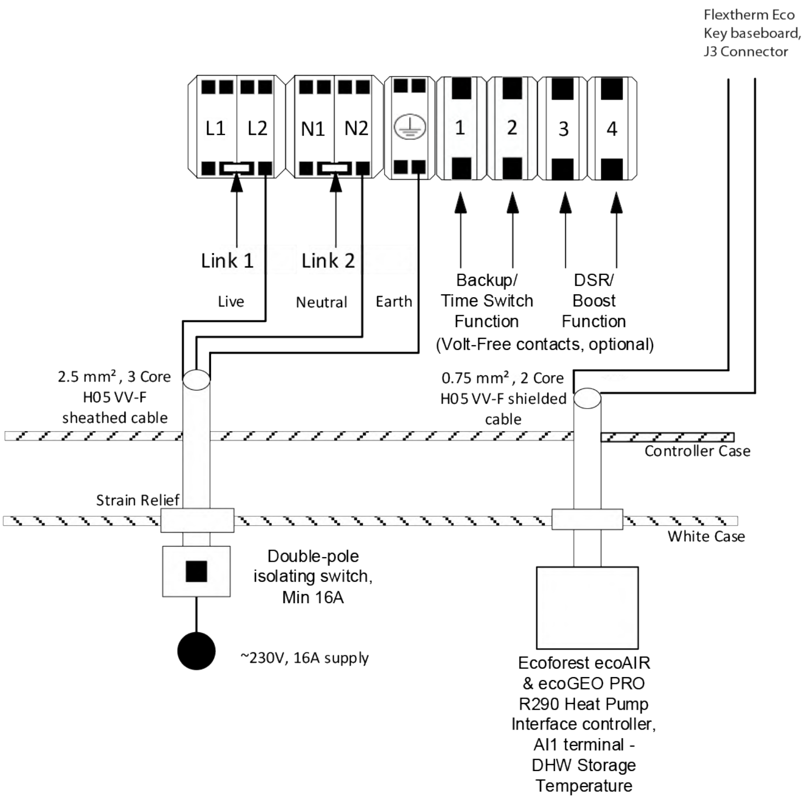

Figure 6: Flextherm Eco G2 D with Ecoforest HP-Key

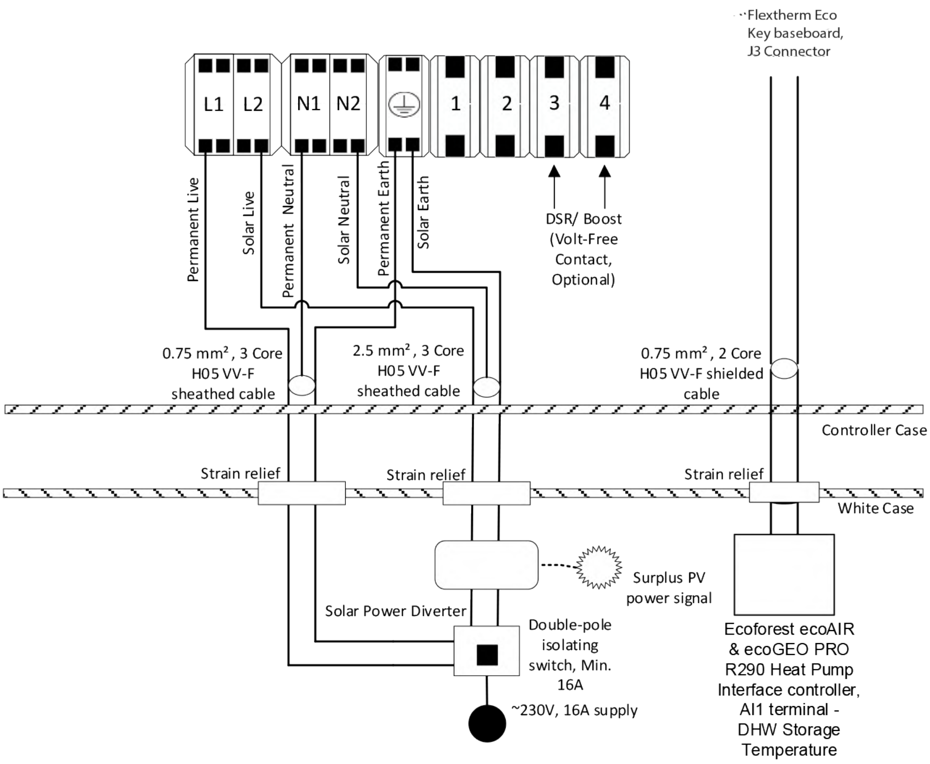

Figure 7: Flextherm Eco G2 D with Ecoforest PV+HP-Key

Notice
For other hydraulic connections and wiring instructions, please follow the instructions of the Flextherm Eco G2 D Installation Manual (https://flamco.aalberts-hfc.com)

6.2. Ecoforest ecoAIR & ecoGEO Controller Settings

On the Main Menu select «DHW/Legionella Protection Menu», then apply the following settings to the parameters detailed below:

Parameter	Settings
DHW	Enable
SetT	60°C
DTstart	5°C
Legionella	Disable

Table 5: Ecoforest DHW and Legionella settings to apply

Notice
If following a DHW daily timed schedule, please ensure that a minimum time window inline with the installed Heat Battery and Heat Pump capacities is chosen, and that potential defrost cycles during this window are accounted for

7. Panasonic L Series R290 Heat Pump

Warning
All Electrical wiring must be carried out by a competent person and be in accordance with the latest local wiring codes and regulations.
Risk of electric shock - potential dual-supply. Always isolate the power supply/ies to the Heat Battery, Heat Pump & Solar Diverter controller (if used) before working on the appliances

When installing the Flextherm Eco G2 D with the heat pump, the steps below should be followed.

These settings are applicable to the following products:

- Flextherm Eco G2 D with the «Panasonic HP-key» & «Panasonic HP+PV-key» (Flextherm Eco G2 D manual - Section 6.4.2 for wiring instructions & Figures 8 & 9 below)
- Heat Pumps from the Panasonic L Series R290 Heat Pumps.

7.1. Wiring

Using a 2-core shielded cable 0.75mm2, the cable will act as a hot water heating tank sensor from the Flextherm Eco Board - terminal J3 into the Heat Pump Indoor Unit controller PCB Terminal - (Tank Sensor) (please refer to HP installation manual). Please run the wire into the heat battery appliance via the appliance case cabling grommets and then into the control box housing through the hole available. Secure the cables in terminals J3 on the Flextherm Eco Board; please see Figures 8 & 9 below for reference.

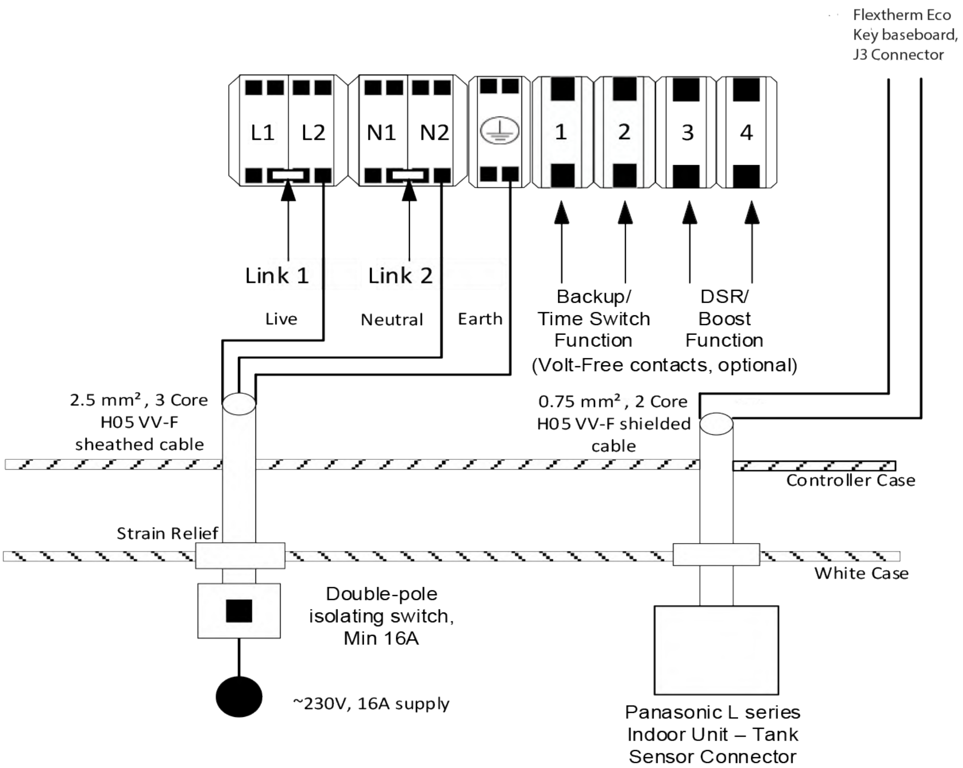

Figure 8: Flextherm Eco G2 D with Panasonic HP-Key

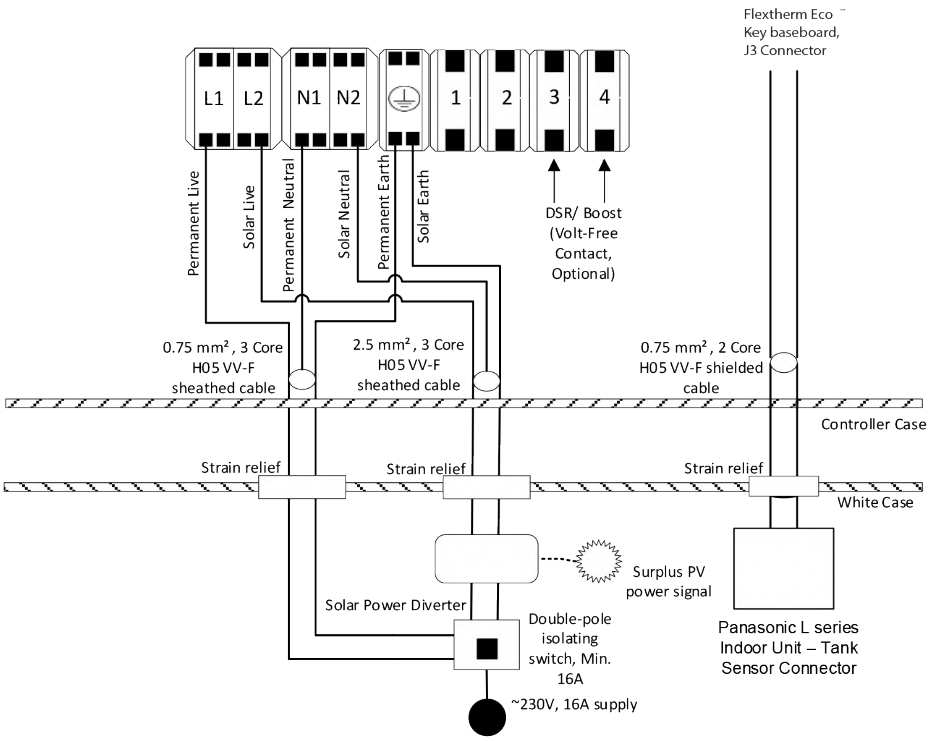

Figure 9: Flextherm Eco G2 D with Panasonic PV+HP-Key

Notice
For other hydraulic connections and wiring instructions, please follow the instructions of the Flamco Flextherm Eco G2 D.

7.2.Panasonic L Series R290 Controller Settings

On the Panasonic L Series R290 Indoor unit controller, the following settings need to be applied:

Parameter	Settings
Tank Connection	Yes
DHW Capacity	Standard
DHW Set Temperature	55 °C
Tank Reheat Temperature	8 °C

Table 6: Panasonic L Series R290 conteoller settings

Notice
If following a DHW daily timed schedule, please ensure that a minimum time window inline with the installed Heat Battery and Heat Pump capacities is chosen, and that potential defrost cycles during this window are accounted for

8. PHNIX GreenTherm R290 Heat Pump

Warning
All Electrical wiring must be carried out by a competent person and be in accordance with the latest local wiring codes and regulations.
Risk of electric shock - potential dual-supply. Always isolate the power supply/ies to the Heat Battery, Heat Pump & Solar Diverter controller (if used) before working on the appliances

When installing the Flextherm Eco G2 D with the heat pump, the steps below should be followed.

These settings are applicable to the following products:

- Flextherm Eco G2 D with the «Phnix HP-key» & «Phnix HP+PV-key» (Flextherm Eco G2 D manual - Section 6.4.2 for wiring instructions & Figures 10 & 11 below)
- Heat pump from the Phnix Greentherm range (R290)

8.1. Wiring

Using a 2-core shielded cable 0.75mm2, the cable will act as a hot water heating tank sensor from the Flextherm Eco Board - terminal J3 into the Heat Pump Interface controller PCB Terminal - (TT- DHW Temp Sensor) (please refer to HP installation manual). Please run the wire into the heat battery appliance via the appliance case cabling grommets and then into the control box housing through the hole available. Secure the cables in terminals J3 on the Flexthem Eco G2 board please see figure 10 & 11 below for references.

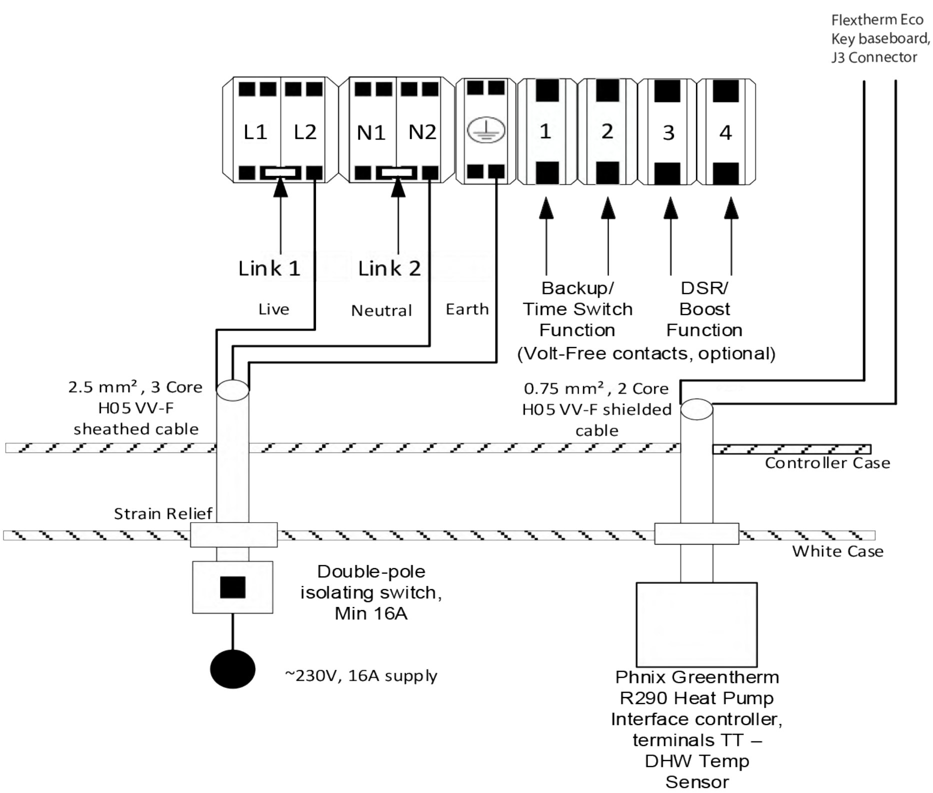

Figure 10: Flextherm Eco G2 D with Phnix HP-Key

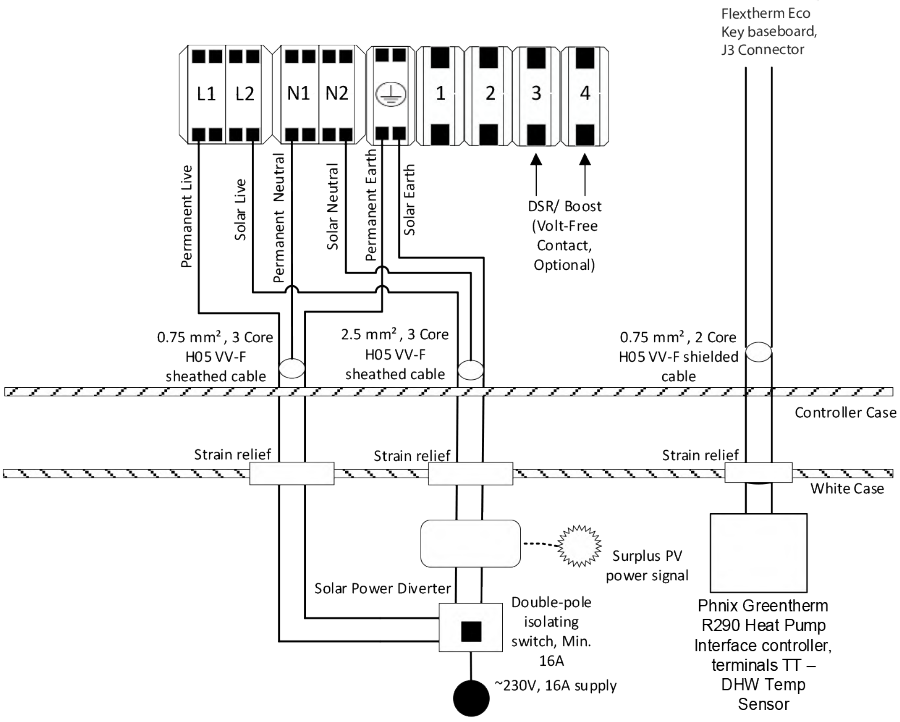

Figure 11: Flextherm Eco G2 D with Phnix PV+HP-Key

Notice
For other hydraulic connections and wiring instructions, please follow the instrucrions of Flamco Flextherm Eco G2 D.

8.2. PHNIX GreenTherm Heat Pump (R290) Controller Settings

- On the Main Menu select «Mode». Select DHW from the Mode menu.
- Jumper the DHW terminals on the Heat Pump wiring terminals «5 & 6», «DHW ON/OFF».
- Unlock the screen to access the «Parameter» Menu. Please apply the following changes to the parameters:

Parameter Reference & Description	Parameter Values
HO1 - AutoStart	YES
R01 - Hot Water Setpoint	55 °C
R16 - Power-on Return Difference of Tank Water	5 °C
R17 - Standby Temp Difference of Tank Water	5 °C
R37 - Max Hot Water Setpoint	70 °C

Table 7: Phnix Greentherm HP controller settings

Notice
If following a DHW daily timed schedule, please ensure that a minimum time window inline with the installed Heat Battery and Heat Pump capacities is chosen, and that potential defrost cycles during this window are accounted for

 We reserve the right to change designs and technical specifications of our products.

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9. Samsung HTQ R32 Heat Pump

Warning
All Electrical wiring must be carried out by a competent person and be in accordance with the latest local wiring codes and regulations.
Risk of electric shock - potential dual-supply. Always isolate the power supply/ies to the Heat Battery, Heat Pump & Solar Diverter controller (if used) before working on the appliances

When installing the Flextherm Eco G2 D with the heat pump, the steps below should be followed. These settings are applicable to the following products:

- Flextherm Eco G2 D with the «Samsung HP-key» & «Samsung HP+PV-key» (Flextherm Eco G2 D manual - Section 6.4.2 for wiring instructions & Figures 12 & 13 below)
- Heat Pumps from the Samsung HTQ range (R32)

9.1. Samsung HTQ R32 Heat Pump

Using the Samsung Tank sensor part supplied with the heat pump, remove the sensor end by cutting off the copper cylinder, and strip the wire as required. The cable will act as a hot water heating tank sensor from the Flextherm Eco key baseboard - terminal J3 into the Heat Pump Interface controller PCB Terminal - (DHW Tank sensor - CNS042) terminal (please refer to HP installation manual). Please run the wire into the Heat Battery appliance via the appliance case cabling grommets and then into the control box housing through the hole available. Secure the cables in terminals J3 on the Flextherm Eco Board, please see Figures 12 & 13 for reference.

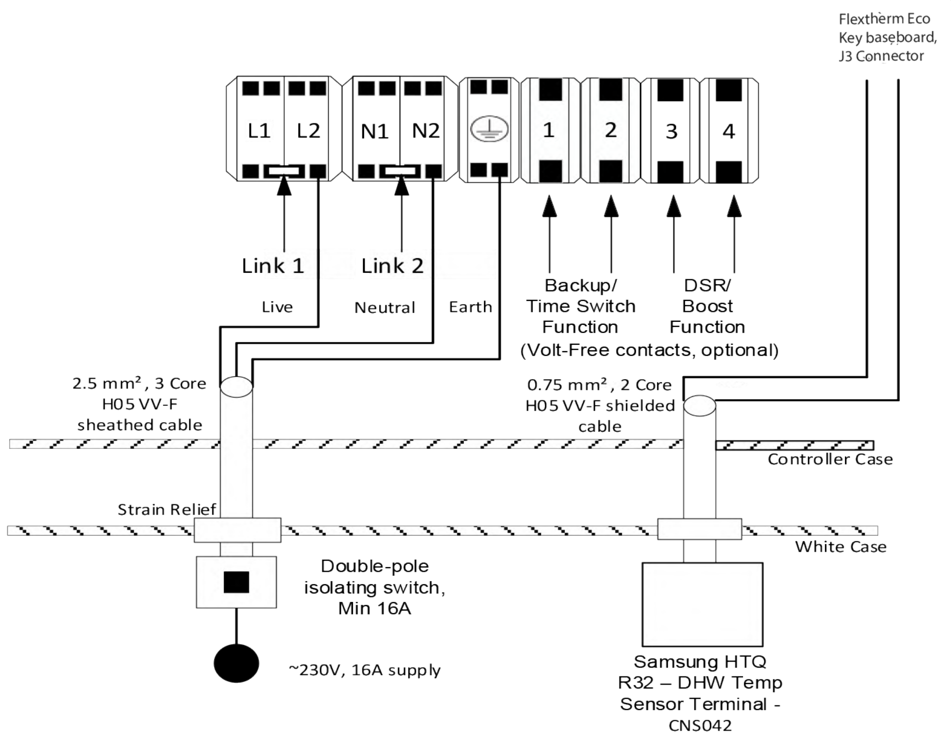

Figure 12: Flextherm Eco G2 D with Samsung HP-Key

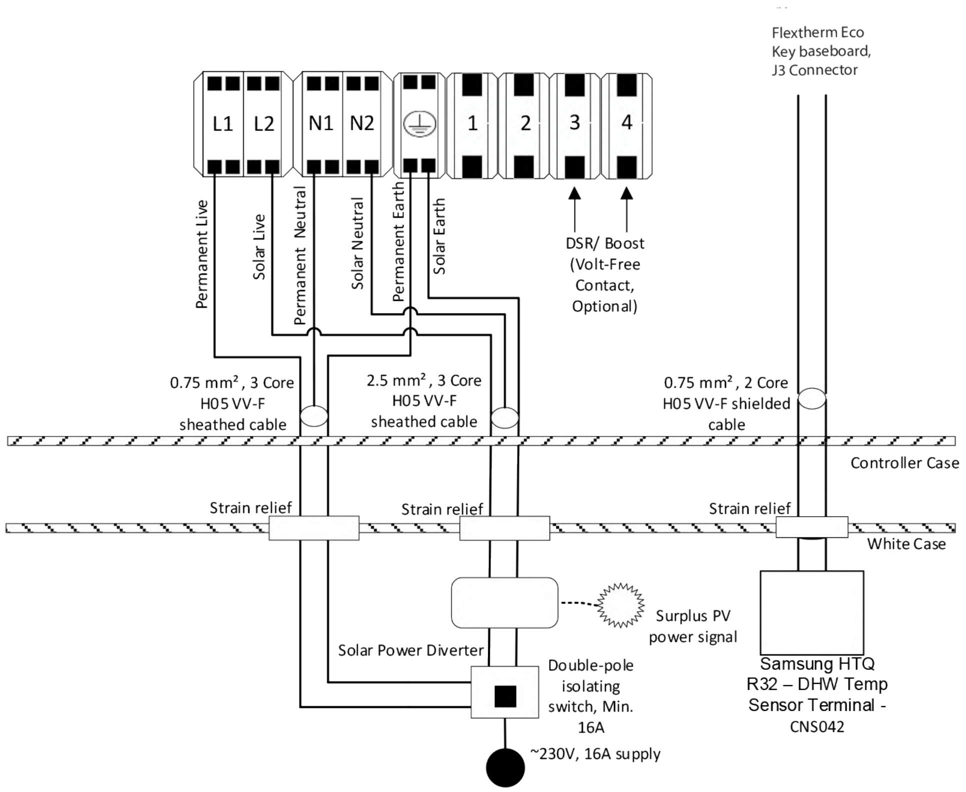

Figure 13: Flextherm Eco G2 D with Samsung PV+HP-Key

Notice
For other hydraulic connections and wiring instructions, please follow the instrucrions of Flamco Flextherm Eco G2 D.

9.2. Samsung HTQ R32 Heat Pump Controller Settings

On the Main Menu of the Heat Pump Controller, ensure «DHW Mode» setting «Standard» is applied.

The following can be accessed in the «FSV» Menu. Please enter the code detailed in the Samsung HTQ R32 manual if required to access the menu

FSV	Parameter Values
3011 - DHW Application	USE
3021 - Max Temp	55 °C
3025 - Max DHW Operation Time	95 mins
1051 - DHW Tank Temperature	70 °C

Table 8: Samsung HTQ R32 HP controller settings

9.3. Using Samsung Kit A 1136

Wire the 2-core PVC insulated cable provided (C2295) from the Samsung Controller Booster Heater terminal connectors «L» & «N» (please refer to Heat Pump manual), run the wire into the relay box (C2291) provided into «TRIGGER INPUT AC» terminals (please refer to relay box instruction sheet). Then wire another 2-core PVC insulated cable provided (C2295) from the relay box terminals «NO1» & «C1» to the Heat Battery, into the control box housing through the opening available. Secure the cables in Terminal 1 & 2 independently. Please note that the polarity of the wires is not important in this wiring setup. Please ensure to use the provided relay backbox (C2292) & 2 x cable grommets (C2296) when running the wires into the relay box

Notice
This function allows the backup heating element inside the Heat Battery to be activated. Please note that this will stop the Heat Battery being charged in Heat Pump mode. This can lead to increased electricity consumption, resulting in higher energy costs. This should be explained to the end user

10. Vaillant Arotherm + R290 Heat Pump

Warning
All Electrical wiring must be carried out by a competent person and be in accordance with the latest local wiring codes and regulations.
Risk of electric shock - potential dual-supply. Always isolate the power supply/ies to the Heat Battery, Heat Pump & Solar Diverter controller (if used) before working on the appliances

When installing the Flextherm Eco G2 D with the heat pump, the steps below should be followed.

These settings are applicable to the following products:

- Flextherm Eco G2 D with the «Vaillant HP-key» & «Vaillant HP+PV-key» (Flextherm Eco G2 D manual - Section 6.4.2 for wiring instructions & Figures 14 & 15 below)
- Heat Pumps from the Vaillant Arotherm + range (R290)

10.1. Wiring

Using a 2-core shielded cable 0.75mm2. The cable will act as a hot water heating tank sensor from the Flextherm Eco Board - terminal J3 into the Heat Pump Interface controller PCB Terminal - (SP1 - screw terminals 1 & 2) (please refer to HP installation manual). Please run the wire into the Heat Battery appliance via the appliance case cabling grommets and then into the control box housing through the hole available. Secure the cables in the terminal J3 on the Flextherm Eco Board; please see Figures 14 & 15 below for reference.

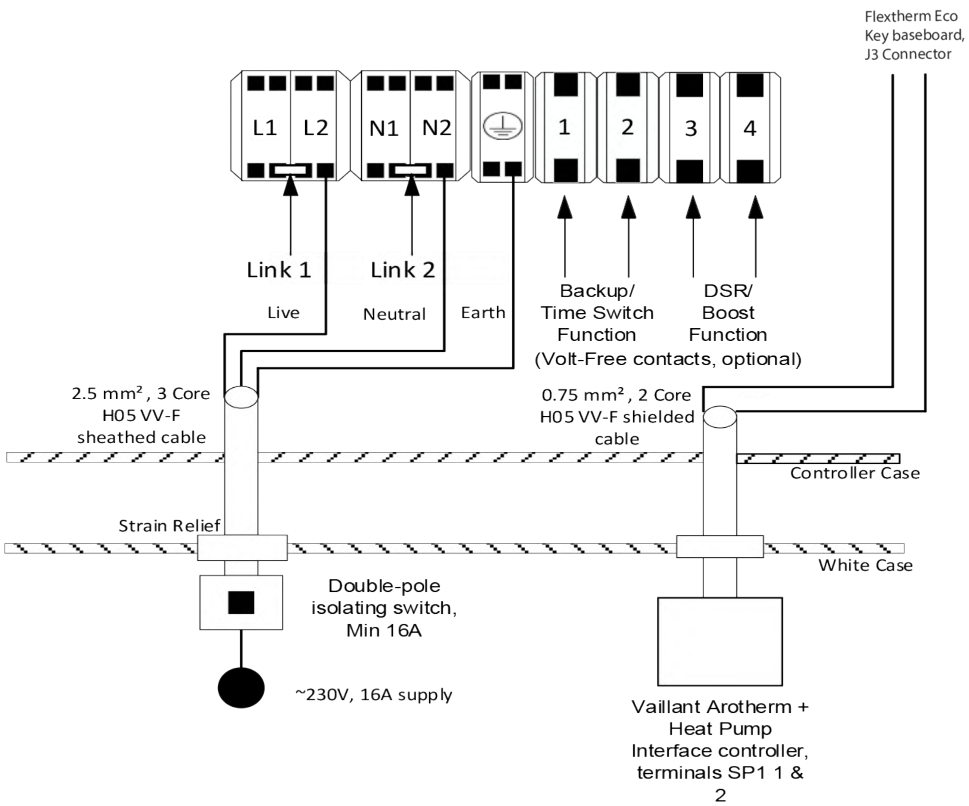

Figure 14: Flextherm Eco G2 D with Vaillant HP-Key

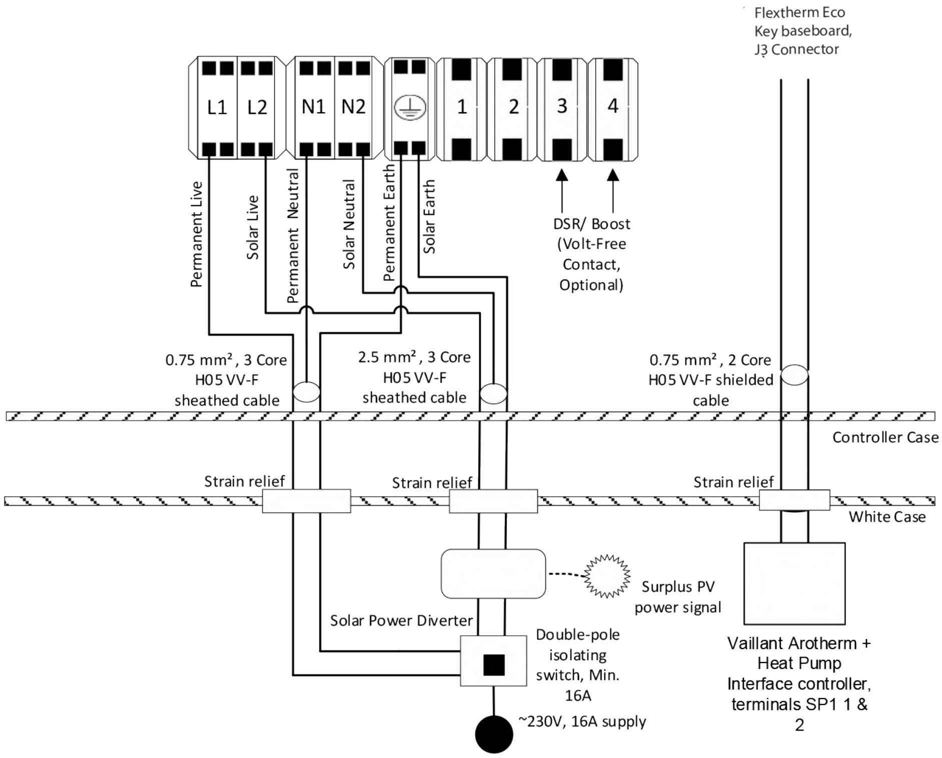

Figure 15: Flextherm Eco G2 D with Vaillant PV+HP-Key

Notice
For other hydraulic connections and wiring instructions, please follow the instructions of the Flextherm Eco G2 D Installation Manual: (https://flamco.aalberts-hfc.com)

10.2. Vaillant Arotherm + R290 Heat Pump Sensocomfort Controller Settings

In the Installer level menu on the Vaillant Sensocomfort controller, please apply the following settings for DHW settings ONLY (please check Vaillant Arotherm Plus Sensocomfort installation and user manual):

Settings	Value required
Max cylinder charging time	120 minutes
Cylinder charging anti-cycling time: min	Minimum setting
DHW target temperature	70°C
Cylinder charging hysteresis	5K
Cylinder charging offset	5K
Anti-legionella. Day & time	OFF

Table 9: Vaillant Arotherm + R290 HP Sensocomfort controller settings

Notice
If following a DHW daily timed schedule, please ensure that a minimum time window inline with the installed Heat Battery and Heat Pump capacities is chosen, and that potential defrost cycles during this window are accounted for

10.3. Vaillant Arotherm + R290 Heat Pump Interface settings

Change Heat pump Compressor output in DHW mode from Eco to Normal. This is available in the DHW settings of the Heat Pump Interface (please check Vaillant Arotherm Plus installation and user manual).

Contact us !

We supply products for the installation industry in more than 70 countries. This is done from Flamco sales offices and via distributors who know the local market, and can give you the right advice at all times.

Aalberts hydronic flow control

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